

Air University



SSDD (Lab)

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Lab Task # 02

Case Study: Security Monitoring Platform (SMP)

Overview

The Security Monitoring Platform (SMP) is designed to monitor and assess the security of software applications. It retrieves source code, performs multi-layered security analysis using external tools, aggregates findings, and presents comprehensive security reports to users. The platform also supports feedback and continuous monitoring to ensure ongoing software security.

External Entities

- **User:** Initiates security assessments, selects software projects, views reports, provides feedback, and schedules future analyses.
- **Version Control System (VCS):** Stores software source code and responds to retrieval requests from the SMP.
- **Static Code Analyzers:** Examine source code to detect security vulnerabilities and coding errors.
- **Text Mining Tools:** Analyze comments, documentation, and textual elements to identify potential security risks.
- **Software Metrics Tools:** Evaluate software quality metrics such as complexity, maintainability, and code coverage.

Functional Processes

1. User Interaction

- The user logs into the SMP, and credentials are verified.
- The user selects a software application for analysis.
- The user initiates the security assessment process.

2. Source Code Retrieval

- The SMP sends a request to the Version Control System for the selected project's source code.
- The VCS returns the source code, which is stored temporarily within the platform.
- The retrieved code is forwarded to the security analysis processes.

3. Security Analysis

- **Static Code Analysis:** The source code is scanned for vulnerabilities and coding flaws, and results are returned to the SMP.
- **Text Mining Analysis:** Code comments and documentation are analyzed for suspicious patterns and security risks.
- **Software Metrics Evaluation:** Metrics such as complexity and maintainability are calculated and returned to the system.

4. Results Aggregation and Reporting

- The SMP aggregates results from all analysis tools.

- A comprehensive security report is generated.
- The report is delivered to the user for viewing or download.

5. Feedback and Continuous Monitoring

- The user provides feedback or requests additional analysis.
- The user schedules periodic security assessments (e.g., daily or weekly).
- The SMP automatically initiates future analyses based on the defined schedule.

Data Flow Summary

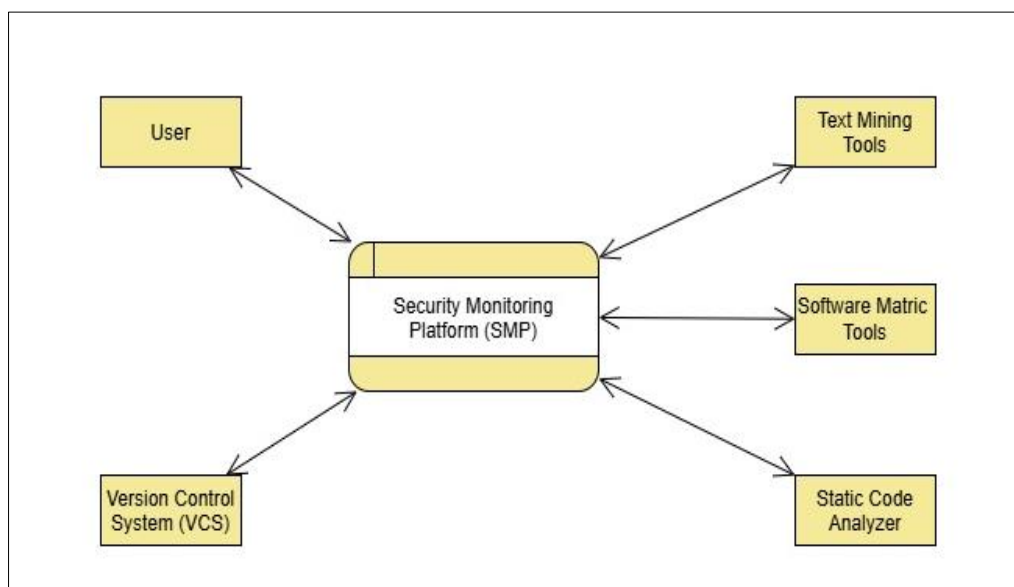
- **Data Sources:** Source code from the VCS; analysis results from Static Code Analyzers, Text Mining Tools, and Software Metrics Tools.
- **Data Stores:** Temporary source code storage, analysis results database, and user interaction logs.
- **Processes:** Data-driven interactions between users, external tools, data stores, and internal system processes.

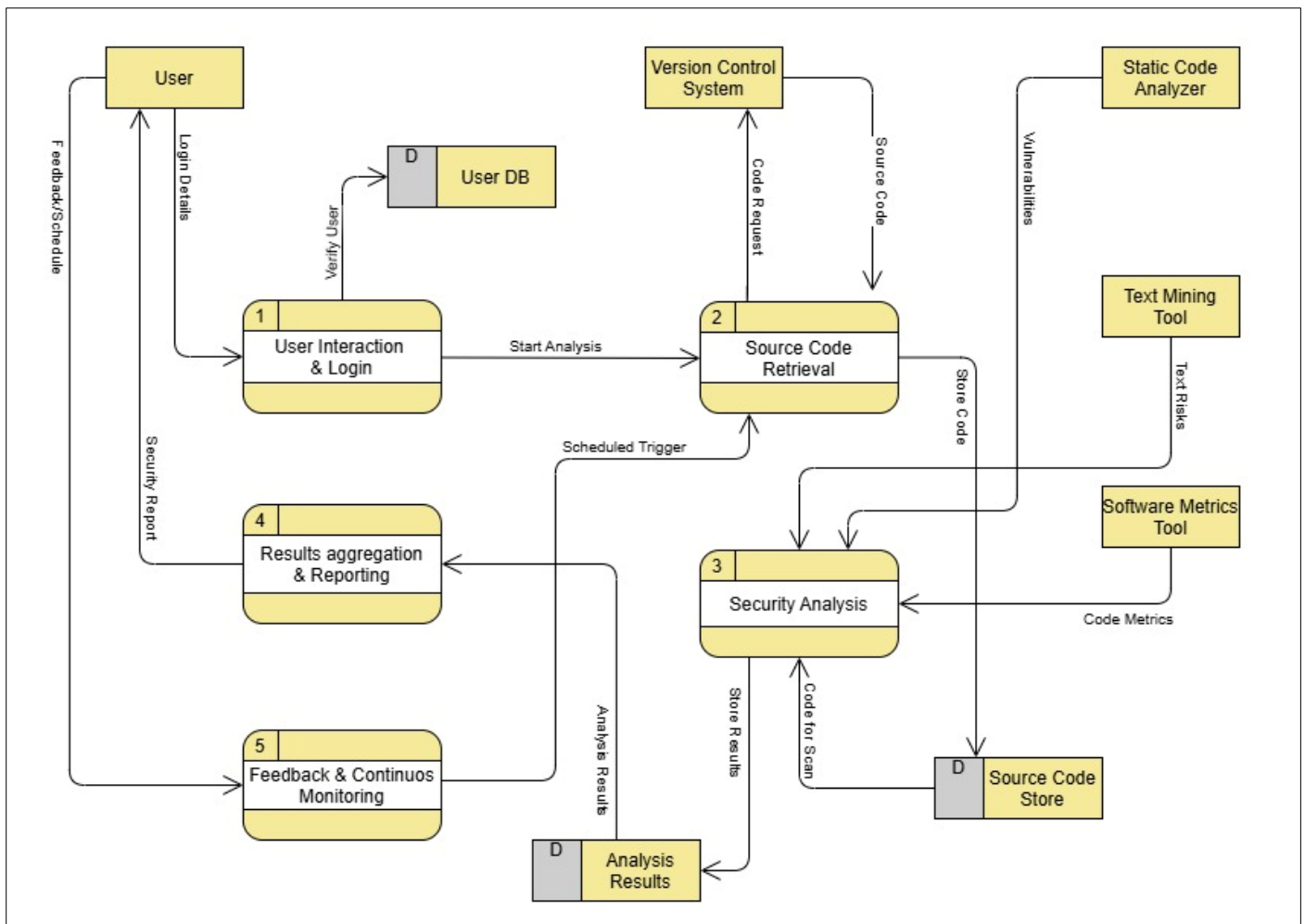
DFD Requirements

- **Level 0 DFD (Context Diagram):** Represents the SMP as a single process interacting with five external entities: User, VCS, Static Code Analyzers, Text Mining Tools, and Software Metrics Tools.
- **Level 1 DFD:** Decomposes the system into the following processes:
 - User Interaction
 - Source Code Retrieval
 - Security Analysis
 - Results Aggregation and Reporting
 - Feedback and Continuous Monitoring

Data Flow Diagrams

Level 0:



Level 1:**THE END**